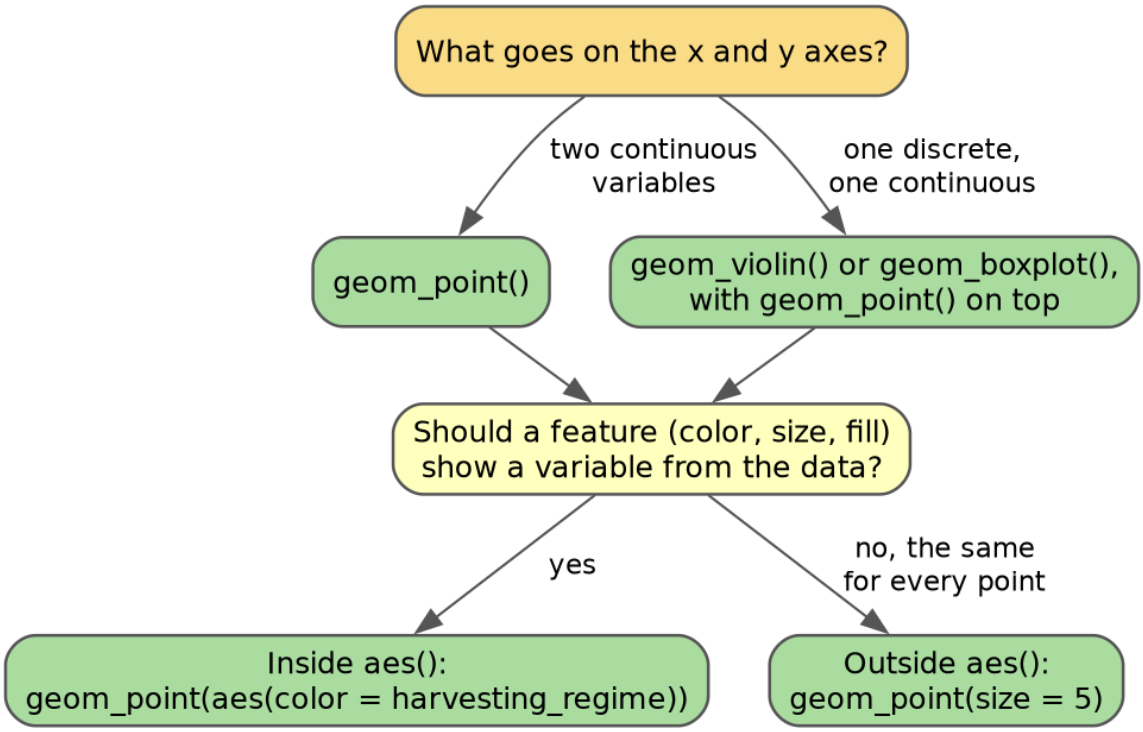
# data visualization I: summary

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Objects store data (like: `my_object <- 7`), and functions act on it: `filter()` keeps only the rows that meet a logical condition, such as `filter(my_data, chemical_species == "FAs")`. A ggplot is then built in three steps: name the data with `ggplot(data = ...)`, map variables onto the axes with `aes()`, and add one or more geoms with `+`. Geoms are geometric objects with features (size, color, shape, etc.). A feature mapped to a **variable** in the data goes **inside** `aes()` (`geom_point(aes(color = harvesting_regime))`), and a feature set to a **constant** goes **outside** `aes()` (`geom_point(size = 5)`). Swapping mappings around is cheap, so it is worth trying a few to see which plot best communicates the aspect of your data that you are interested in.

| Function or operator | What it does | What we vary |
|---|---|---|
| `<-` | stores a value or a result in an object | the object name on the left |
| `filter()` | keeps only the rows that meet a condition | the condition: `==`, `!=`, `<`, `<=`, `>`, `>=`, `%in% c(...)` |
| `ggplot()` | starts a plot and names the data | `data =` |
| `aes()` | maps variables onto the axes and onto features of the shapes | `x`, `y`, `color`, `size`, `fill` |
| `+` | adds a geom to the plot, and several geoms can be stacked | end the line with + to continue on the next |
| `geom_point()` | draws one point per observation | `aes(color = ...)`, `aes(size = ...)`, or constants such as `size = 5`, `color = "black"` |
| `geom_violin()` | draws the spread of values in each group | `aes(fill = ...)` |
| `geom_boxplot()` | draws a box plot for each group | nothing extra in this chapter |